
EE 501: Power System Operation and Control

Course Code: EE 501

Course Name: Power System Operation and Control

L-T-P-C: 3

Prerequisite: Consent of the faculty member

Students intended for: UG and PG

Elective or Core: Elective

Approval: 2nd Senate

Course Contents

An overview of power system operations and control, basic objectives of security and economics in power system operation and control, security analysis, reactive power dispatch and optimal power flows, automatic generation control, reactive power control, state estimation, brief introduction to power system restructuring and power market operations.

Text & Reference Books:

Textbooks:

1. A. J. Wood and B. F. Wollenberg, **Power generation, Operation and Control**, 2nd Edition, John Wiley and Sons, 1996.
2. Prabha Kundur, **Power System Stability and Control**, Tata Mcgraw Hill Education Private Limited, 2006.
3. Loi Lei Lai, **Power System Restructuring and Deregulation: Trading, Performance and Information Technology**, John Wiley & Sons, 2001.